Specification

Product Name: 6 LEDs 175° Matrix LED Module DC12V

Product Model: PO-MB8361-W

CE	RoHS	LED Matrix	DC 12V	2835 SMD	IP 65	5 Years

Product Model

PO-MB8361-W 68 × 68 × 8.3 mm 6 LEDs SMD2835 175° waterproof constant voltage DC12V matrix LED module

Picture

Features

1. This product uses high-efficiency 2835 LED beads, achieving a luminous efficacy of 120 lm/W.
2. The product uses a wide-angle 175° lens, ensuring uniform light distribution.
3. The product uses an aluminium substrate, resulting in good heat dissipation.
4. The product uses PVC injection moulding and an epoxy resin coating process, resulting in aesthetically pleasing and lightweight modules.
5. CE and RoHS certificated, ensuring reliable quality.

Applications

Suitable for 8–15 cm illuminated signs and light boxes; used in single-sided light boxes for advertising signage in airports, subways, banks, buildings, shopping malls, etc.

Parameters

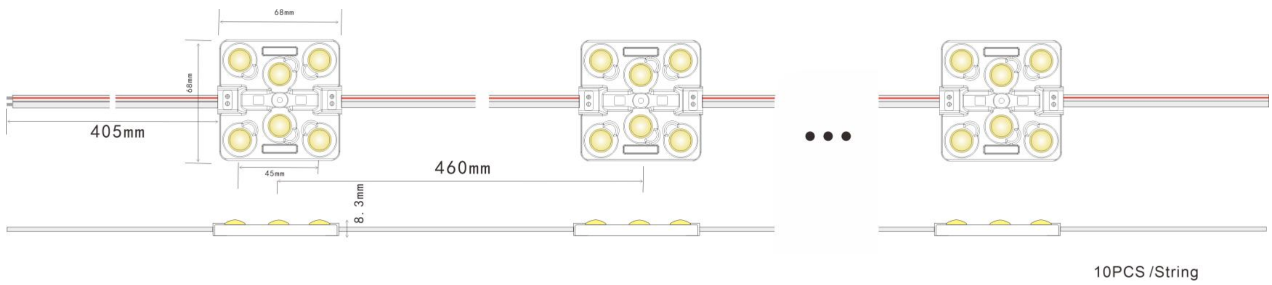
Product Code	LED Colour	Colour Temp. (K)	CRI	SD CM	Beam Angle (°)	Wavelength (nm)	Luminous Flux (lm)	Efficacy (lm/W)	Voltage (VDC)	Current (mA)	Power (W/pcs)
PO-MB8361-W	White	3000 / 4000 6500 / 10000	≥80	≤5	175	660–720	110–120	≥80	12	500	6

Other Parameters

Product Code	IP Rank	Operating Temp. (°C)	Storage Temp. (°C)	Standard Cascade (pcs)	Single-ended Max Cascade (pcs)	Double-ended Max Cascade (pcs)	Module Weight (g/pc)
PO-MB8361-W	IP65	-25 ~ +60	-25 ~ +70	10	10	20	53

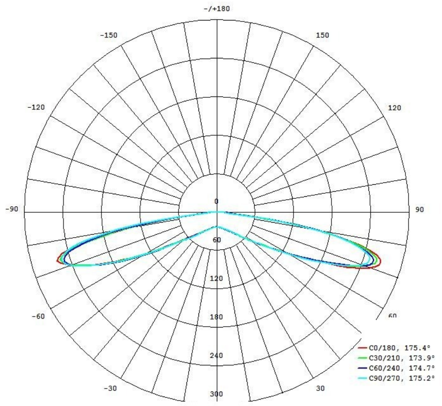
Remarks: (1) Test environment temperature is 25 ± 2°C. (2) The above data are typical values; actual product parameters may differ from the typical data and are subject to change without prior notice. (3) "—" indicates the parameter is not required for the time being.

Dimensions



10 pcs / string

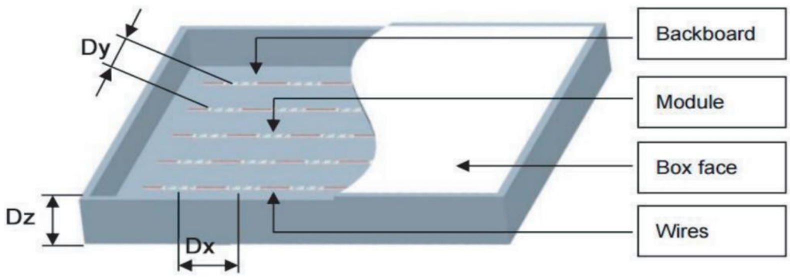
Light Distribution Curve



Polar intensity distribution — C0/180: 175.4°, C30/210: 173.9°, C60/240: 174.7°, C90/270: 175.2°. Average beam angle (50%): 174.8°.

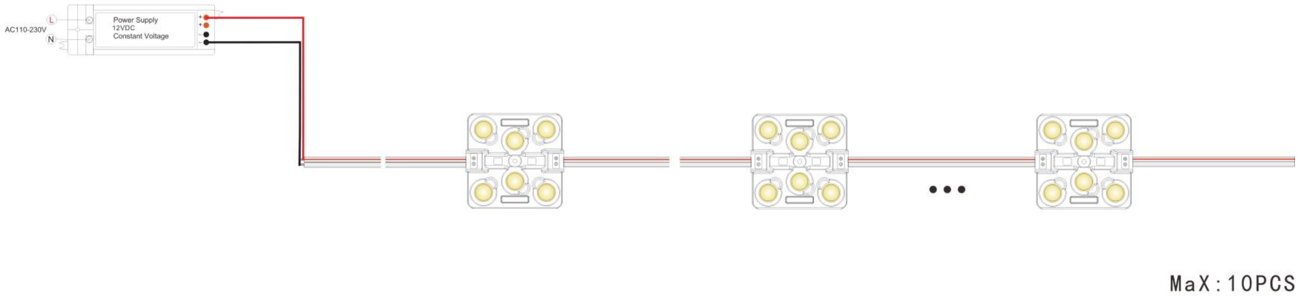
Typical Lighting Test Data

Light Box Thickness (mm)	Module Spacing (mm)	Number of Modules (pcs/m²)	Surface Illuminance (Lx)	Power per m² (W/m²)
80	Dx=280, Dy=280	14	4500–5000	84
100	Dx=360, Dy=360	9	1600–1800	54
120	Dx=460, Dy=460	4	1200–1300	24
150	Dx=460, Dy=460	4	1000–1100	24



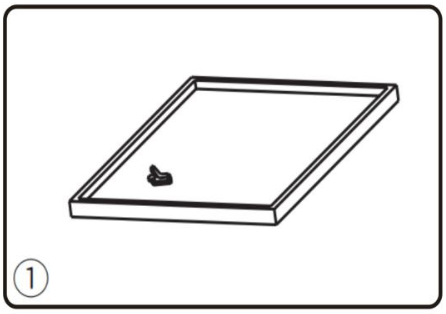
Light box construction: backboard, module, box face and wires (Dx / Dy module spacing, Dz box depth).

Critical Connection Diagram

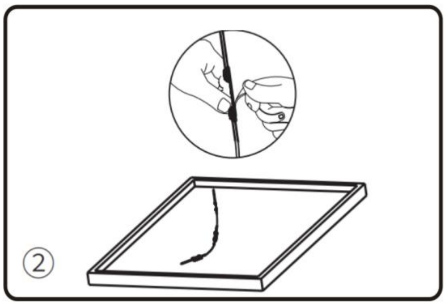


AC110–230V input to 12VDC constant voltage power supply. Maximum 10 pcs per string.

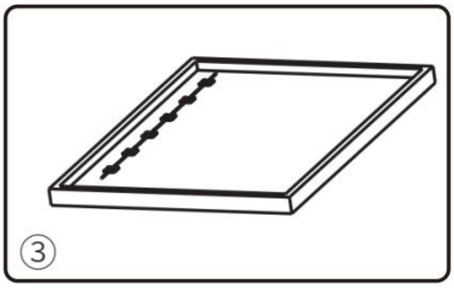
Installation Steps



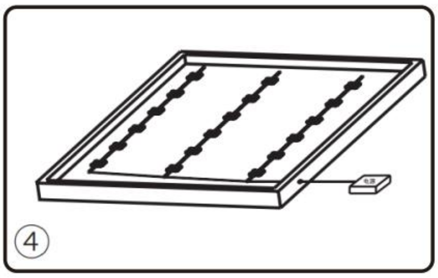
Step 1. Clean up any debris on the mounting surface.



Step 2. Test the layout: refer to the lighting data to determine module spacing and position, peel off the double-sided adhesive release paper on the back of the module, lightly stick it to the mounting back plate and screw it in for positioning.



Step 3. Route the main power cord on one side of the light box and connect the module interfaces to the main cord, ensuring positive and negative terminals are correctly connected. Adjust the number of cascaded modules per the reference value.

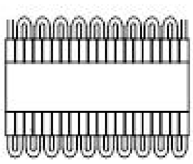


Step 4. Connect the power supply to the main line inlet of the light box, check that the wiring is correct, then test by powering it on.

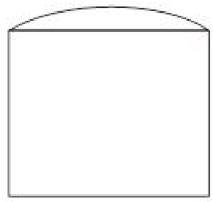
Packaging Information

Product Code	Per String (pcs)	Quantity (bags)	Total Quantity (pcs)	Total Weight (kg)	Outer Box L (mm)	Outer Box W (mm)	Outer Box H (mm)
PO-MB8361-W	10	40	400	16.5	520	300	360

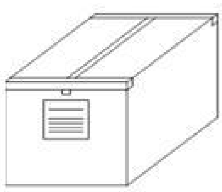
Packaging Diagram



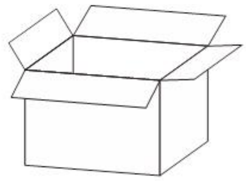
10 pcs / string



Anti-static bag, 10 pcs / bag



Sealing



400 pcs / carton

Common Product Failures & Troubleshooting

Fault Phenomenon	Possible Cause	Solution
All lights off	No electricity supply	Restore power transmission.
	The switching power supply auto-protects when its output terminal is open or short-circuited.	Troubleshoot the fault and re-energise.
	Module power supply polarity is reversed	Check the wire connections to ensure the positive and negative poles are correct.
Some lights off	Some switching power supplies are not powered	Check the power supply system and troubleshoot.
	Some module power supply lines are faulty	
	Polarity of some modules is reversed	Correct the wiring.

Common Product Failures & Troubleshooting (continued)

Fault Phenomenon	Possible Cause	Solution
LED brightness varies or is not bright enough	Power supply overload	Replace with a higher power-rated supply according to the load.
	Switching power supply line loss is too large or uneven	Ensure module operating voltage is within ±5% of rated voltage: (1) shorten the power cord between supply and module or use larger-diameter wire; (2) ensure the number of modules per channel is within the maximum allowed cascade and keep cascade counts similar across channels.
	Too many modules connected in series	Adjust the number of modules per power supply branch so each circuit stays within the maximum allowable cascade requirement.
LED flashing	Poor contact at the wiring point	Find the poor-contact point and eliminate the fault.
	Switching power supply failure	Replace the switching power supply.

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